

Portfolio deployment guide

Docker, Caddy and shared administrator login

Updated 8 October 2026 (Asia/Kolkata). This guide records the completed Oracle setup and the commands used to operate it. It is not a live status monitor.

Resource	Current state
Public portfolio	https://rithvik.ddns.net/ - deployed, SMTP configured
Admin dashboard	https://rithvik.ddns.net/admin/ - stage 1 deployed
Shared sign-in	https://rithvik.ddns.net/auth/ - Authelia username/password
Oracle checkout	/opt/portfolio on main; server IP 144.24.135.201
Portainer	Stage 2 - awaiting approval; not installed
Blog + learning notebook	Stage 3 - awaiting separate approval; not implemented

How requests reach the application

```
Browser -- HTTPS --> Caddy -- public request --> Node :4173
|
+-- login/session check --> Authelia :9091
+-- verified identity ----> Node /admin/
```

No-IP maps the hostname to Oracle. Caddy publishes ports 80 and 443 and obtains the HTTPS certificate. App and auth ports stay private inside Docker. HTTP visits to the IP redirect to the HTTPS hostname.

The dashboard links to the portfolio and shows inactive cards for future applications. It does not yet manage containers or edit articles. Adding a card does not grant access or configure SSO.

Verified for stage 1

15 automated tests passed, along with production page/link checks and live HTTPS authentication, logout and access-protection checks. Desktop and phone layouts were reviewed locally. Contact availability was checked without sending email.

Files, images and local preview

A Dockerfile builds an image. Compose defines how containers use that image, their settings, networks, ports and storage. Build creates the image; up creates or updates containers; up --build does both.

File	What it controls
Dockerfile	Builds and tests the Node application. Runtime uses Node 24.
compose.yaml	Local public-site preview at http://localhost:8080 .
compose.ip.yaml	Temporary HTTP preview; no SMTP or shared login.
compose.production.yaml	Current app, Caddy and Authelia production stack.
.dockerignore	Allowlist of build inputs; excludes credentials and private data.
deploy/Caddyfile.docker	Current production HTTPS and route configuration.
deploy/authelia/configuration.yml	Non-secret login, session and access rules.
deploy/Caddyfile + portfolio.service	Alternative host/systemd deployment; not active on Oracle.

Local Docker Desktop

```
docker compose config --quiet
docker compose up --build -d --wait
docker compose ps
# Open http://localhost:8080
docker compose logs --tail=100 app
```

Run in the repository root with Linux containers. Host port 8080 maps to app port 4173 inside the container. A running container does not mean host port 4173 is published. Rebuild after edits; this preview has no hot reload.

The optional local .env supplies SMTP settings. Without them, the contact form offers LinkedIn. Caddy and Authelia are not part of local Compose; /admin/ stays closed. The separate direct Node preview uses <http://127.0.0.1:4173>.

Use one Compose file per deployment. The IP and production files are standalone, not overrides. Stop the IP preview before using production because both need port 80.

Update the running Oracle site

1. Commit development changes locally. Regenerate public pages when content/templates changed. **2.** Merge V2 into main and push main. **3.** Connect to Oracle over SSH and run:

```
bash /opt/portfolio_redeploy.sh
```

The shortcut already exists and calls `/opt/portfolio/portfolio_redeploy.sh`. Run this inside SSH, not in Windows PowerShell. You do not need to create another script or paste its contents. A Git push alone does not deploy.

What the script does

Locks against concurrent deployment; fetches main; permits a fast-forward only; validates Compose, Authelia and Caddy; builds the app and generated pages; runs checks/tests; starts healthy services; reloads Caddy; verifies public and protected endpoints.

The current site serves during the build. App replacement can briefly interrupt requests. Authelia is recreated and its in-memory sessions expire, so sign in again. Certificates, database and private files persist.

Initial copied files and future commits

Stage 1 was copied to Oracle before a Git commit. Include the matching implementation and documentation in the next merge. The script stashes server copies only when they exactly match origin/main. If it reports a difference, reconcile that file before retrying; do not discard private settings.

Environment-only changes

```
bash /opt/portfolio_redeploy.sh --no-pull
```

This rebuilds the existing checkout without fetching Git. It uses the server `.env.production`. Editing your local `.env` does not update Oracle. A plain Docker restart does not rebuild code or apply new Compose environment settings.

Read-only checks inside SSH

```
cd /opt/portfolio
# Optional shorthand for this SSH session:
dc() { sudo docker compose --env-file .env.production \
  -f compose.production.yaml "$@"; }
dc ps
dc logs --tail=100 app caddy auth
curl -I https://rithvik.ddns.net/
curl -I https://rithvik.ddns.net/admin/
```

A signed-out admin request should redirect to `/auth/`. The `dc` helper above is only a shell shortcut; it does not create a file or change a service.

Administrator access and recovery

Open <https://rithvik.ddns.net/admin/> and sign in as **rithvik**. The initial password is in **Portfolio admin login.txt** beside the Oracle SSH key on your computer. Its server copy is `/opt/portfolio/.auth/bootstrap-login.txt`. Keep it private and save it in your password manager.

Session behavior

Sessions expire after 30 minutes of inactivity or 8 hours total. Remember-me is disabled. Sign out ends the current session; restarting Authelia ends all its in-memory sessions. This stage uses a username and password. TOTP, WebAuthn, email reset and other application SSO clients are not configured.

Why login headers cannot be forged

Caddy removes visitor-supplied Remote-* identity headers, asks Authelia to verify the session, and forwards verified claims. Node checks the actual proxy peer IP, enabled/trust settings, the exact configured username and admins membership. Dashboard responses are no-store and noindex. Authelia defaults to deny outside its configured access rules.

Change or recover the password in SSH

Back up the user file privately. Generate an Argon2id digest interactively, entering the new password at the prompt rather than on the command line:

```
cd /opt/portfolio
sudo docker compose --env-file .env.production \
  -f compose.production.yaml exec auth \
  authelia crypto hash generate argon2 --parallelism 1
```

Edit `.auth/secrets/users.yml` with nano. Replace only the quoted password hash with the generated digest. Preserve the username, admins group, and valid JSON-compatible YAML syntax. Then restart auth:

```
sudo docker compose --env-file .env.production \
  -f compose.production.yaml restart auth
```

Verify the new login and update your password manager. The bootstrap file and local login file are not updated automatically. Do not remove `.auth/` or replace the database encryption key to reset a password.

Future SSO integrations

The identity service is ready, but each future application still needs a supported integration. Forward authentication does not automatically remove an application's own login. Portainer will be configured only after stage 1 approval; the writing tools follow after stage 2 approval.

Caddy routes and new sites

Edit `deploy/Caddyfile.docker` for the running Docker stack. Caddy reads it at `/etc/caddy/site/Caddyfile.docker`. The bare `deploy/Caddyfile` is for the unused `host/systemd` alternative.

Route	Current destination
<code>/</code> and public pages	Node app:4173
<code>/api/contact</code>	Node contact handler with same-origin submission rules
<code>/auth/</code> and its endpoints	Authelia auth:9091, with the path preserved
<code>/admin/</code> and its subpaths	Authelia authorization, then Node app:4173

Adding another hostname

Create its DNS record pointing to Oracle. Start the application on a deliberate network shared with Caddy and keep its host port unpublished. Add a top-level Caddy site block. Example for a **public** app; replace all example values:

```
tool.example.com {
    encode zstd gzip
    reverse_proxy tool:3000
}
```

Use the container service name and internal port. `localhost` inside Caddy refers to Caddy itself. Do not attach unrelated apps to the trusted portfolio backend. A private app also needs access rules and identity integration; the example above has no login protection.

Adding a path on the existing hostname

A path such as `/tool/` needs no new DNS record or wildcard. The application must support that public base path. Inside the existing hostname block, before its fallback handle:

```
redir /tool /tool/ 308
handle_path /tool/* {
    reverse_proxy tool:3000
}
```

`handle` preserves the path; `handle_path` strips the prefix, so `/tool/page` reaches the upstream as `/page`. Asset links and redirects must still work. These examples are documentation only; no tool service, Portainer or blog route has been added.

Proxy changes and domain moves

Caddy-only edits already saved on Oracle

Validate before reloading. Run the second command only if validation succeeds:

```
cd /opt/portfolio
sudo docker compose --env-file .env.production \
  -f compose.production.yaml exec -T caddy \
  caddy validate --config /etc/caddy/site/Caddyfile.docker \
  --adapter caddyfile &&

sudo docker compose --env-file .env.production \
  -f compose.production.yaml exec -T caddy \
  caddy reload --config /etc/caddy/site/Caddyfile.docker \
  --adapter caddyfile
```

For changes to Compose, DOMAIN, or auth settings, use `portfolio_redeploy.sh` so affected containers receive the new environment and auth configuration. Caddy and Authelia mount configuration directories, so Git file replacement is visible.

Move to a purchased domain

1. Point the new hostname to Oracle. **2.** Change DOMAIN in server `.env.production` to the hostname only. **3.** Run the existing-checkout redeployment:

```
bash /opt/portfolio_redeploy.sh --no-pull
```

Compose derives the public origin, build `SITE_URL`, and Authelia URLs from DOMAIN. Caddy requests the new certificate. Preserve storage and sign in on the new hostname afterward. Future OAuth/OIDC callback URLs must be updated too.

Why the hostname still matters after DNS

DNS finds the server IP. TLS and HTTP still carry the hostname so Caddy can choose the certificate and site. The Node public origin controls contact submissions; Authelia uses the hostname for cookies and redirects. This deployment redirects HTTP on the IP and is not configured for raw-IP HTTPS. IP certificates are not categorically impossible.

Keep the No-IP hostname renewed according to your account requirements. Update its record if Oracle's IP changes. No Cloudflare Pages, Worker, tunnel, or dynamic DNS updater is deployed by this setup.

Private settings and backups

Location	Preserve it for
<code>.env.production</code>	Domain, SMTP credentials, proxy and account settings
<code>.auth/secrets/users.yml</code>	Account metadata, groups, Argon2id password hash
<code>.auth/secrets/session-secret</code>	Session secret
<code>.auth/secrets/storage-key</code>	Encryption key required by the auth database
<code>.auth/data/</code>	SQLite database and filesystem notifier output
<code>.auth/bootstrap-login.txt</code>	Initial plaintext credential record; keep private
<code>portfolio_caddy_data</code>	Certificates and ACME account state
<code>portfolio_caddy_config</code>	Caddy runtime state

Keep private files out of Git, Docker build context and public uploads. `.auth/` directories use mode 700 and secret files use mode 600, owned by the deployment user. Auth runs as UID/GID 1000:1000 on this server. Its data is writable; its secrets are read-only inside the container.

Make a consistent auth backup

Briefly stop only auth, archive the complete `.auth/` directory with permissions, then start auth again. The portfolio can stay up while sign-in is unavailable. Store the backup privately with `.env.production`; preserve both Caddy named volumes separately. Ordinary rebuilds and `compose down` preserve this data; `down --volumes` deletes named volumes.

Restore the storage key and database together. Do not regenerate encryption keys while keeping the old database. The one-time setup helper refuses to overwrite `.auth/` for this reason.

Existing stage 1 recovery points

Private server directory: `/home/opc/.config/portfolio-deploy/admin-stage1-YzXVP4Gu/`. It contains `source-before.tar.gz` and `auth-stage1.tar.gz`. The previous app image is tagged `rithvik-portfolio:before-admin-stage1`. These are manual recovery points, not scheduled backups.

A rollback needs compatible source, Compose/Caddy configuration and app image together. Preserve current files before restoring; changing the app image alone does not undo proxy or auth changes. See `deploy/ADMIN.md` for the full recovery record.

First setup, checks and references

On a new Linux server only

Install Docker Engine/Compose, Git, Python 3, curl and flock. Clone main into /opt/portfolio, configure DNS and firewall ports 80/443, then create .env.production from deploy/production.env.example. Use mode 600 and your server UID/GID values. Run as the normal deployment user:

```
cd /opt/portfolio
python3 scripts/setup-admin.py
bash portfolio_redeploy.sh --no-pull
```

Already completed on the current instance. The initializer generates a private account and keys, and refuses to overwrite .auth/. The /opt shortcut is specific to the existing Oracle setup; the repository script works on a new server.

Symptom	Check
Local 4173 is empty	Docker local preview uses http://localhost:8080.
SSH key not accessible	Use the full quoted path to the key on this computer. Copying a key does not change its identity.
Public site times out	DNS, running Caddy, Oracle ingress and host firewall.
Admin redirect to /auth/	Expected when signed out; local preview has no auth stack.
502 on login or admin	Auth container health, mount permissions, network and logs.
Logged out after redeploy	Expected: auth sessions live in memory and auth restarts.
Server files differ from main	Commit/merge matching changes or reconcile the named file.
Contact available is true	SMTP is configured; verify actual inbox receipt separately.

Source documents and reference links

Repository: README.md, deploy/DOCKER.md, deploy/ADMIN.md, and deploy/README.md. The last file is the alternative systemd guide, not the active Docker workflow.

Caddy path handling - preserving or stripping a path prefix.

Authelia with Caddy - authentication at the reverse proxy.

Authelia sessions and private secrets.

Verification limits: live authentication was checked with HTTPS requests; the in-app browser could not open the live hostname during review. Layout was reviewed locally. No test email was sent. Portainer and writing tools remain uninstalled pending their separate approvals.